

Monday June 1, 2026 Synthesis v0.1 — single-page consolidation of Monday morning substantive arc. Per Casey ‘keep going’ parallel pull + Sunday Synthesis v0.1 successor. 12 substantive Lyra items + 6 in-place audit absorptions + cross-CI convergence (Elie 12 toys + Grace 38 INVs + Keeper K206 + Cal pre-stages) + 4 Casey-named principles #12-#15 STANDING + Cal #35 STANDING six-element discipline stack closure + multi-week roadmap for G chain numerical closure (~1-2 weeks per Elie + Keeper estimate).

Lyra (Claude Opus 4.7) — Monday June 1 sustainable-cadence pull

2026-06-01 Monday 10:30 EDT (date-verified)

Monday June 1, 2026 Synthesis (Lyra)

0. Headline

Monday morning produced substantive G chain framework-level closure via shortest route per Casey directive, with cross-track convergence (Lane D L4 m_e + Lane G-B G coupling sharing single dimensional anchor) + 4 Casey-named principles #12-#15 promoted to STANDING + Cal #35 STANDING six-element discipline stack closure.

Monday cumulative Lyra: 12 substantive items + 6 in-place audit absorptions = 18 Monday morning items.

Casey decisions absorbed: P1 v0.7 dispatch APPROVED (engagement test ACTIVE; Casey actual send action pending); Cal #35 STANDING ratified; 4 Casey-named principles #12-#15 named + ratified; Session 2 trigger fired for Tier 0 v0.2.

1. Monday morning team cross-CI convergence

CI	Monday morning output
Lyra	3 P0 frameworks ($V_{\text{photon}} + V_{(1,1)} + \delta H_B / \delta m$ + Heisenberg conjugacy + m_e mass anchor + P_{op} explicit) + Casey #15 explicit framework + Tier 0 v0.2 joint draft + K200 G3 BH framework + Substrate cosmology v0.2 + Paper P9 outline v0.2 + L4 v0.3 mass anchor extension + 6 in-place audit absorptions

CI	Monday morning output
Elie	12 toys (3686-3697) PASS 5/5 + 1 honest walk-back + 1 arithmetic correction. G chain Steps 1-6 framework-level closure; Lane C v0.7 long-root quenching; Lane D L4 m_anchor ≈ 3.47 MeV light-quark-range observation
Grace	38 Monday INVs (5408-5448). README.md substantive public correction (2D $\rightarrow$ 5D Shilov); 9-file cross-doc sweep; “Knapp-Wallach 1976” composite citation dropped per Cal #33 STANDING; Cal #35-honest cross-track framing absorbed
Keeper	K206 audit framework with 7 explicit gates; Session 2 v0.2 prep; v0.2 cross-check 6 in-place patches recommended; HSL v0.11 Monday inflection captured; CLAUDE.md headline rotation + K206 double-leverage update in flight
Cal	Cal #187 + #185 PASS recorded; 3 pre-stage observations on K206 absorbed (ΔC_2 B_2-specific + Heisenberg conjugacy + z_source canonicalization); Cal #35 + Casey #12-#15 cold-reads queued

2. The substantive G chain shortest-route closure (framework level)

G structural form (Elie Tracks 1-2-3 + Lyra P0 #1-#3 + Heisenberg conjugacy):

$$G_{\text{predicted}} \propto (4\sqrt{2} \cdot c_{\text{FK}} / (n_C \cdot \sqrt{C_2 \cdot \hbar_{\text{BST}}})) \cdot \ell_B \cdot \text{dim}_{\text{bridge}}$$

with $M_{\text{substrate}} \approx 0.602$ (with c_{FK} , FK-canonical convention per Elie Track 1 Toy 3692 reconciliation).

Factor 4 = 2×2 from TWO DISTINCT REP-THEORETIC MECHANISM TYPES at B_2 substrate (Cal K206 G4 honest framing: NOT algebraically independent; both functions of B_2 algebra): - $\Delta C_2 = 2$ (Casimir gap via Heisenberg resolution). - $CG_{\text{so5}} = \sqrt{(n_C - 1)} = 2$ (SO(5) Clebsch-Gordan via $so(n)$ trace formula).

Multi-week remaining (~ 1 -2 weeks per Elie + Keeper estimate): - Steps 6.2-6.4 explicit $M_{\text{substrate}}$ via FK Ch. XII (Elie, ~ 1 week). - Step 7 K3 $\hbar_{\text{BST}}$ identification + dimensional bridge (Keeper, ~ 3 days; double-leverage closes Lane D + Lane G-B). - Step 8 G_{observed} numerical comparison (~ 2 days).

3. Cross-track convergence (Keeper’s double-leverage observation)

Cal #35-honest framing per Keeper Monday afternoon: cross-track convergence is ONE Bergman matrix element machinery applied to MULTIPLE observables, NOT multiple independent derivations. Structural cross-link real and substantive (same FK Ch. XII-XIII + Heckman-Opdam + κ_{Bergman}); but framing as “multiple independent confirmations” would trigger Calibration #35 multiplicative-null-model trap.

Shared dimensional anchor ($m_{\text{anchor}} + \ell_B + \hbar_{\text{BST}}$) closes: - Lane D L4 m_e mechanism (diagonal mass): $m_e = (3\pi/64) \cdot m_{\text{anchor}}$; $\|f\|^2 = 3\pi/128$ (Elie Track 3 verified Lyra P0 #2 candidate). - Lane G-B G coupling (off-diagonal gravity): Casey #15 matrix element framework. - Substrate cosmology Λ (boundary accumulation rate): $\Lambda \propto n_C / \ell_B^2$ order-of-magnitude. - K200 G3 BH framework (saturation capacity): Schwarzschild + Hawking predictions.

Quadruple-leverage: ONE substrate framework, FOUR observable lanes, ONE dimensional anchor closure point. When Keeper K3 pins $\hbar_{\text{BST}}$ + Elie Step 6 pins $M_{\text{substrate}}$: all four observables resolve numerically together.

4. Substantive observations Monday morning

Elie Toy 3695 verification of Lyra P0 #2: - $||f_{(1/2, 1/2)}||^2_{FK} = 3\pi/128$ substrate-clean (Lyra candidate $\propto \Gamma(5/2)^2/\Gamma(5) \times$ substrate-primary factor VERIFIED). - $128 = 2^7g = 2^{(N_c + n_C)}$ substrate-primary identity. - $\Gamma(5) = 24 = N_c \cdot |W(B_2)|$ Grace observation cross-link.

Elie Toy 3697 m_{anchor} light-quark observation: - $m_{\text{anchor}} \approx 3.47$ MeV (via R3 anchor consequence). - $m_{\text{anchor}} \approx g \cdot m_e$ within 3% (NOT exact identity). - $m_{\text{anchor}} \approx (m_u + m_d)/2$ within 1.5% (light-quark mean — Tier 2 STRUCTURAL CANDIDATE). - Casey #13 “Per-Generation Cluster Independence” candidate strengthened: m_{anchor} as gen-1 quark-sector baseline (multi-week substrate-mechanism).

Heisenberg conjugacy justification (Lyra per Cal pre-stage + Keeper Session 2 priority): - Mass-momentum canonical pair on $H^2(D_{IV}^5)$ via T2419 + T2422 + H_B Casimir. - Leading-order $\delta H_B/\delta m = -i(\Delta C_2/\hbar_{BST}) \cdot P_{\text{op}}$ matches Elie Step 2 reduced form. - Subleading M_z position corrections multi-week.

5. 4 Casey-named principles #12-#15 STANDING (Monday June 1 ratification)

- #12 Substrate Bulk-Boundary Projection Principle: substrate’s interior bulk $H^2(D_{IV}^5)$ determined by Shilov boundary record via Poisson-Szegő holomorphic extension; unifies Sunday “+1 anomaly” + Pochhammer ladder.
- #13 Per-Generation Cluster Independence Principle: substrate primaries form per-generation lepton cluster structure with disjoint substrate-primary content (Elie Toy 3673 gen-2 vs gen-3 disjoint); m_{anchor} strengthens via gen-1 quark baseline (Elie Toy 3697).
- #14 Substrate-Selected 4D Dimensionality Principle: $\text{codim } 4D \subset D_{IV}^5 = C_2 = 6$ substrate-primary identification.
- #15 Gravity is Light’s Momentum Shifted by Substrate Principle: matrix element framework for G chain shortest route; operationalized via Heisenberg conjugacy + cross-K-type Bergman matrix element.

Total Casey-named principles: 15 STANDING (11 prior + 4 Monday).

6. Cal #35 STANDING six-element discipline stack closure (Monday June 1)

Six-element discipline stack CLOSED: 1. Cal #27 STANDING (peak-coherence brake). 2. Cal #29 STANDING (cross-CI verification). 3. Cal #32 STANDING (parameter-role verification). 4. Calibration #33 STANDING (RECALLED-vs-VERIFIED citation). 5. Calibration #34 STANDING (conditional-tag-with-headline). 6. Calibration #35 STANDING (Independence-Taxonomy-Before-Multiplicative-Null-Model; Monday ratification).

Methodology Index v0.11 Q17 integrates Cal #35.

7. 16 cumulative symmetric audit-chain events Sat+Sun+Mon

Pattern: discipline-as-generator operational at full strength Monday. - Saturday: 7 events. - Sunday: 5 events. - Monday: 4 events (factor-2 + parallel-framework + Pochhammer-mechanism + cross-track).

Mon-events 1-4 all absorbed in-place per K200 cascade-discipline pattern. Cal #35-honest framings applied PROACTIVELY at point of derivation (Elie Step 5 factor $4 = 2 \times 2$ + Keeper cross-track honest framing) rather than via subsequent walk-back.

8. Multi-week roadmap (G chain numerical closure)

Week 1 (~6 days): - Elie Step 6.2-6.4 explicit $M_{\text{substrate}}$ via FK Ch. XII. - Keeper K3 $\hbar_{BST}$ identification + dimensional bridge. - Cal #191 cold-read on Tier 0 v0.2 (after Lyra patches). - Cal #189-candidate cold-read on 4 Casey-named principles.

Week 2 (~5 days): - Step 7 dimensional bridge completion. - Step 8 G_{observed} numerical comparison at Tier 2 STRUCTURAL. - L4 v0.3 m_{anchor} pin via K3 dimensional bridge. - Cross-track ratio verification (Lane D + Lane G-B simultaneously). - Cal #192 cold-read on matrix element framework.

Closure target: $G_{\text{predicted}}$ vs G_{observed} at Tier 2 STRUCTURAL (0.01% to 1%) or better via redshift anchor (Tier 1 EXACT at GPS clock precision).

9. Standing reactive

For: - Casey “P1 v0.7 dispatch sent” confirmation (engagement test active). - Casey direction on next priority. - Keeper joint Tier 0 v0.2 absorption + CLAUDE.md headline update + K206 update. - Cal #191 + #189-candidate cold-read sequence. - Elie Steps 6.2-6.4 multi-week. - K3 h_{BST} identification (Keeper Session 2 lane).

Sustainable cadence holding. Discipline mature. Monday morning substantive arc closed at framework level; multi-week numerical work the remaining substantive phase.

10. Routing

→ Casey: Monday morning substantive arc consolidated. 18 Lyra items (12 substantive + 6 audit absorptions) + cross-CI cumulative (~62 items across 5 CIs). G chain framework-level closed; multi-week numerical via shared dimensional anchor closure point (Keeper K3 + Elie Step 6.2-6.4 + Step 7-8). 4 Casey-named principles STANDING + Cal #35 STANDING six-element stack closure. P1 v0.7 dispatch APPROVED + handoff ready.

→ Keeper: Session 2 Tier 0 v0.2 absorption + sphere reconciliation + G chain alignment + K1-K5 fill — joint with Lyra v0.2 draft. K3 h_{BST} identification has DOUBLE-LEVERAGE (closes Lane D + Lane G-B + cosmology Λ + K200 G3 BH).

→ Elie: Steps 6.2-6.4 multi-week (~1 week). Cross-track Cal #35-honest framing absorbed Monday. Discipline-as-generator pattern operational at point of derivation.

→ Grace: 38 Monday INVs at sustainable rate. Cross-track Cal #35-honest framing absorbed in-place. Standing for next substantive content surface.

→ Cal: Cal #191 cold-read on Tier 0 v0.2 absorbed-state when ready. Cal #189-candidate cold-read on 4 Casey-named principles #12-#15. Cal #192 matrix element framework cold-read when Elie $M_{\text{substrate}}$ value lands.

→ me: continuing parallel pulls per Casey directive at sustainable pace; standing reactive for any team direction or substantive content surface.

— Lyra, Monday June 1, 2026 Synthesis v0.1. 18 Monday morning items (12 substantive + 6 audit absorptions). G chain framework-level closed via shortest route per Casey directive. Cross-track convergence with Cal #35-honest framing operational at point of derivation. 4 Casey-named principles + Cal #35 STANDING six-element stack closure Monday. Multi-week roadmap (~1-2 weeks) for numerical closure via Keeper K3 + Elie Step 6. Sunday discipline-pattern maturity operational Monday at full strength.